

Guangrui Li

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EDUCATION

University of Technology Sydney

Ph.D. in Computer Science

Sydney, Australia

July 2019 – May 2024

Chongqing University

B.E. in Software Engineering

Chongqing, China

Sep. 2014 – June 2018

RESEARCH INTEREST

- **Agentic AI:** I have architected and deployed multi-agent system in production (Amazon Rufus). Now I am focusing on devising scalable and efficient agent systems, especially the self-evolving ability of agents and its robustness under continuous environment change, through reinforcement learning.
- **Computer Vision:** I have strong background in computer vision and machine learning, *i.e.*, 2D / 3D segmentation and detection, pose estimation, and Out-of-distribution (OOD) generalization of deep learning models.

PROFESSIONAL EXPERIENCE

Amazon Rufus AI

Applied Scientist II

Palo Alto, CA

06/02/2025 – Now

- Architected the multi-agent routing system from the ground up in Rufus shopping agent, devising effective and scalable MAS in production.
- Investigate how to evolve agent environments in a controllable and scalable manner, develop an Environment Scaling Methodology, and submit it to ICML. Now focus on RL strategies to drive agent under evolving environments.
- Monetization of Rufus: 1) launch the "buy for me" feature in Rufus, delivering +147M annualized OPS; 2) devise and derive sponsored following-ups based on online conversation context in Rufus.

AWS AI Rekognition

Research Intern on Vision Foundation Models

Seattle, WA

11/06/2023 – 02/09/2024

- Worked on enhancing the content moderation services in AWS.
- Focused on improving the robustness and generalizabilities during the fine-tuning of vision foundation models (CLIP, DINOv2, etc), striking the balance between on in-distribution (ID) recognition and robustness to out-of-distribution (OOD) samples.
- Based on the internship project, first-authored a paper that latterly was accepted by ECCV 2024.

Sony AI Research

Research Intern on Vision Foundation Models

Remote

08/21/2023 – 10/30/2023

- Studied Federated Knowledge Distillation for visual foundation models, *i.e.*, gathering and distilling detection knowledge from multiple distributed detectors.
- Collaborated with a team of researchers to develop and integrate a all-in-one vision foundation models.
- Implemented novel techniques for effective class aggregation in federated knowledge distillation.

Baidu Research

Research Intern on Multi-Modal Learning

Beijing, China

Sep 2021 – Aug 2022

- Developed novel methodologies for open-vocabulary detection to identify and localize novel/unseen objects based on language prompts.
- Achieved state-of-the-art results on COCO and LVIS benchmarks, first-authored a research paper.
- The developed approach has been examined and deployed in Baidu Services.

PUBLICATIONS

1. The World Won't Stay Still: Programmable Evolution for Agent Benchmarks
International Conference on Machine Learning (**ICML**), 2026 [Under review].
Guangrui Li, Yaochen Xie, Yi Liu, Ziwei Dong, Xingyuan Pan, Tianqi Zheng, Jason Ingyu Choi, Michael J. Morais, Binit Jha, Shaunak Mishra, Bingrou Zhou, Chen Luo, Monica Xiao Cheng, Dawn Song.
2. Mixture of Prototypes for Test-time Adaptive Segmentation
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
Guangrui Li, Zhengyu Zhu, Yongxin Ge.
3. RAFT: Reasoning-Aware Fine-Tuning for MLLMs
International Conference on Learning Representations (**ICLR**), 2026. [In submission]
Jinhao Li, Guangrui Li, Riheng Zhu, Zhenbang Sun, Feng Liu.
4. Exploring the Role of Explicit Temporal Modeling in Multimodal Large Language Models for Video Understanding
Yun Li, Zhe Liu, Yajing Kong, Guangrui Li, Jiyuan Zhang, Chao Bian, Feng Liu, Lina Yao, Zhenbang Sun
5. Robustness Preserving Fine-tuning using Neuron Importance
European Conference on Computer Vision (**ECCV**), 2024.
Guangrui Li, Rahul Duggal, Aaditya Singh, Kaustav Kundu, Bing Shuai, Jonathan Wu.
6. Construct to Associate: Cooperative Context Learning for Domain Adaptive Point Cloud Segmentation
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024.
Guangrui Li.
7. Decouple to Contrast: Orthogonalized Ambiguity Reduction for Open-Vocabulary Object Detection
Guangrui Li, Yifan Sun, Zongxin Yang, Errui Ding, Yi Yang
8. Adversarially Masking Synthetic to Mimic Real: Adaptive Noise Injection for Point Cloud Segmentation Adaptation
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023.
Guangrui Li, Guoliang Kang, Xiaohan Wang, Yunchao Wei, and Yi Yang
9. Decompose to Generalize: Species-Generalized Animal Pose Estimation
International Conference on Learning Representations (**ICLR**), 2023.
Guangrui Li, Yifan Sun, Zongxin Yang, and Yi Yang
10. VSPW: A Large-scale Dataset for Video Scene Parsing in the Wild
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2021.
Jiaxu Miao, Yunchao Wei, Yu Wu, Chen Liang, Guangrui Li, Yi Yang
11. Domain Consensus Clustering for Universal Domain Adaptation
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2021.
Guangrui Li, Guoliang Kang, Yi Zhu, Yunchao Wei, and Yi Yang
12. Content-Consistent Matching for Domain Adaptive Semantic Segmentation
European Conference on Computer Vision (**ECCV**), 2020.
Guangrui Li, Guoliang Kang, Wu Liu, Yunchao Wei, and Yi Yang

ACADEMIC SERVICES AND AWARDS

Google Travel Grant, 2023
ICLR financial assistance award, 2023
Semi-finalist in Valeo International Challenge [24/1376], 2017
Reviewer of IEEE TPAMI, TKDE, TIP, and TNNLS.
Programme Committee of CVPR, ECCV, ICCV, ICLR, ICML, and NeurIPS.

SKILLS

Programming: Python (Pytorch, vLLM, LangChain), C/C++, Shell, L^AT_EX.
Languages: Native in Chinese (Mandarin), Fluent in English.